

Music Player Database



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- Database to play music for users
- Users manage their accounts & playlists
- Retrieve information about account
- AI generate/edit playlists for users based on their listening statistics
- GUI for users to interact with database easily with simple format

What Is Our Project?

Database

- Expandable
- Easy to edit
- Procedures for easy access
- 3rd Normal Form
- Satisfying Functional Dependencies

5 Java Workflows

- 2 simple workflows
- 3 more complex workflows
- Business logic
- Provide methods for the GUI

User GUI

- User friendly
- Simple
- Easy to use

Designs & Tests

- Diagrams for project design
- (Activity & Sequence diagrams)
- Unit tests for java methods
- Unit tests for database stored procedures

Requirements

User Creates a Playlist

User Input

- User ID & Desired Playlist Name

Business Layer

- Takes input, calls stored procedure in database with input

Stored Procedure

- Takes the input of a User ID & Playlist Name to initialize the playlist into the database

Workflow 1

Display Playlist Information

User Input

- User ID & Playlist Name

Business Layer

- Calls upon stored procedure to get a user's listening stats
- Displays these stats to user

Stored Procedure

- Takes User ID to get the user's song statistics

Workflow 2

- User Inputs their ID
- Then, they input 3 artist preferences
- These artists are used to create the playlist

1. User Input

- Songs from the artists are assigned a score
- Score determined by skips, plays, last played, etc.

2. Song Recommendations

- Songs never played before are given a score boost
- Promotes users hearing new songs by liked artists
- Similar to other music platforms

3. Never Played Songs

- This forms a list of recommended songs based on score
- Song list is returned

4. Returned Song List

Auto Playlist Builder

This is more complex in the business logic layer than previous workflows, making it the first Nontrivial Workflow

Allows a user to clean up their playlist, getting rid of disliked songs with a song scoring system

- Searches through all of user's songs in playlist, checks all requirements to the right
- Adds to List: 'SongsToRemove' if disliked

Playlist Purge

Skips

- Removed if song has been skipped too much
- Removed if song has too high of a skip:play ratio

Listens

- Removed if song is not listened to
- Removed if not listened to in X months

Genre

- Removed if the genre is not popular among user's liked songs

Artist

- Removed if the artist is not popular among user's liked songs

Blacklist

- Removed if the song is on the users blacklist

User Input

- Two user IDs
- Playlist from each user that they want to combine
- New Playlist's name

Song Scoring

- Similar scoring to other Nontrivials
- Skips, Last Played, Listens
- If both users have a high score on a song, it is chosen

Playlist Creation

- All chosen songs are added to new playlist
- Uses input's playlist name
- Both users have access to this playlist now

Joined Playlist

GUI

Music Player Database



Scroll Feature

Scroll through playlist statistics in the Playlist Stats window, accessed through the home screen drop down menu

Text Fields

Does not showcase password logins for database connection, and instead displays them as symbols



Code Demo Here



Demonstration

Thank You!

**Any
Questions?**